BEFORE THE WORKSHOP · READ ME FIRST

Come prepared: your setup checklist

From Excel to Python and AI · a two-day workshop for finance students · University of Chicago

Total time: about 45 minutes
plus ~60 min of pre-reading
Finish **before Day 1**

The workshop runs over two days — **Day 1**, a Python bootcamp, and **Day 2**, an AI day joint with UChicago tech majors, ending in a team build sprint. Everything runs in your web browser — **nothing needs to be installed**. You need a laptop, a Google account and a GitHub account. The steps below get all three working and prove it. Do them in order, before Day 1. Written for people who have never coded; if you have, it takes fifteen minutes.

What to bring on the day

- Laptop (Mac or Windows) **and charger** — both days
- Your Google login and your GitHub login
- Chrome or Edge (Safari and Firefox work; Chrome is smoothest)
- A personal laptop if your work laptop blocks websites or downloads

1 Run your first Python program in Google Colab — 10 min

Colab is a free notebook that runs Python on Google's computers. It is what we use all day, so this is the one step that matters most.

1. Go to colab.research.google.com and sign in with a Google account. **Use a personal Gmail** if your university account says Colab is disabled.
2. Click **File → New notebook**. A page with an empty grey box (a "cell") opens.
3. Click in the cell and type the line on the right, exactly, including the quotes and parentheses.
4. Press **Shift + Enter** to run it. The first run takes ~10 seconds while Colab connects.
5. Click the title ("Untitled0.ipynb") at the top, rename it **hello.ipynb**. It autosaves to your Google Drive.

```
print("Hello, world!")
# after Shift+Enter you should see:
Hello, world!
```

✔ **Expected: the words Hello, world! appear under the cell**

Red text instead? Compare your line with the one above character by character — a missing quote or parenthesis is the usual cause. Fix it and press Shift+Enter again.

2 Confirm the packages we need are there — 3 min

A package is a bundle of ready-made Python tools. Colab comes with ours pre-installed; this cell just proves it.

1. In the same notebook, hover under your first cell and click **+ Code** to add a new cell.
2. Type (or paste) the four lines on the right and press **Shift + Enter**.
3. Three version numbers appear, one per line. The exact numbers do not matter.

If a package is ever missing, one line installs it: `!pip install package-name` (note the exclamation mark). We do this once in class.

```
import pandas, numpy, matplotlib
print(pandas.__version__)
print(numpy.__version__)
print(matplotlib.__version__)
2.2.2
2.0.2
3.10.0   # your numbers may differ
```

✔ **Expected: three version numbers, no red text**

3 Create a GitHub account and your first repository — 10 min

GitHub stores code with its full history and lets teams work on it together. In the last hours of Day 1 your group will push code there and watch it be tested automatically.

1. Go to github.com/signup. Choose the **Free** plan. Pick a username you would be happy to put on a CV — you will share it with your group.
2. Verify your email (check spam). GitHub will ask you to set up **two-factor authentication**; choose the phone-app option and finish it now.
3. Already have an account? Just sign in and skip to step 4.
4. Top-right, click the **+** then **New repository**. Name it **hello-workshop**, keep it **Public**, tick **Add a README file**, click **Create repository**.
5. Click the pencil icon on the README, type a sentence about yourself, click **Commit changes…** then **Commit changes**. That was your first commit — a saved, named snapshot.
6. **Now connect Colab to GitHub** (the exact move we use in class): back in your Colab tab, click **File → Save a copy in GitHub**. Authorise Colab when asked, choose the repository **hello-workshop**, leave the rest as is, click **OK**.
7. Refresh your repository page. **hello.ipynb** is now listed next to the README — you have just "pushed" your first code.

✔ **Expected: github.com/your-username/hello-workshop shows README.md and hello.ipynb**

Optional: GitHub Desktop is a friendly app for the same actions from your laptop. Not needed for the workshop.

4 Run the workshop's setup check 5 min

One notebook that loads the workshop data and tells you whether you are ready. If it says so, you are.

1. **Click here to open 00_setup_check in Colab.** (Or on the repository: notebooks → 00_setup_check.ipynb → orange **Open in Colab** badge.)
2. Click **Runtime** → **Run all**. If Colab warns the notebook *was not authored by Google*, click **Run anyway** — normal for GitHub notebooks.
3. Wait ~30 seconds, then scroll to the bottom and look for the message on the right.

```
Loaded 236,549 rows x 12 columns
You're ready ✔  See you at 9:00.
```

✔ **Expected: "You're ready" at the bottom of the notebook**

Remember that row count: Excel stops at 1,048,576 rows; the real file has hundreds of millions.

5 Get a free AI assistant account — your debugging partner 5 min

When code shows red text, the fastest fix is to paste the error into an AI assistant. You will do this all day.

1. Sign up for the free tier of any one: claude.ai, chatgpt.com or gemini.google.com.
2. Test it. In Colab run `print(Hello)` — no quotes, on purpose. You get `NameError: name 'Hello' is not defined`. Copy that red text.
3. Paste it into the assistant with: *"I'm a beginner learning Python. I ran this in Google Colab and got this error. Explain what went wrong and how to fix it."* ✔ **It should tell you to put quotes around Hello**

Never paste passwords, API keys or confidential work documents into any AI chat tool. Error messages and public data are fine.

6 Pre-reading and a short survey ~60 min, at your own pace

1. **Python basics (30 min).** *Python for Data Analysis*, ch. 2, §2.1–2.3, free at wesmckinney.com/book/python-basics. Skim; the workshop repeats it.
2. **The data (10 min).** Look up your home address at broadbandmap.fcc.gov. Note the providers and speeds listed.
3. **How AI models work (10 min).** 3Blue1Brown, *But what is a GPT?* — 3blue1brown.com/lessons/gpt. Watch at least to the "next-token prediction" section.
4. **GitHub Hello World (10 min).** docs.github.com → Hello World: branches and pull requests in the browser.
5. **Survey (2 min).** A few quick questions, including your coding background and GitHub username. Fill it in after step 4: **forms.gle/12uzGqjYnx3BtkD88**

Recommended if you have time: Anthropic's prompt engineering overview (15 min).

+ Optional: Python on your own laptop 20 min · Colab is all we use in class; this is for coding afterwards without a browser

1. Install **Python 3.12** from python.org/downloads. **Windows:** tick **"Add python.exe to PATH"** on the first installer screen. **Mac:** just run the installer.
2. Install **VS Code** from code.visualstudio.com; open it, click the Extensions icon (four squares), search **Python**, install the one by Microsoft.
3. **Terminal** → **New Terminal**, then: `python --version` (Mac: `python3 --version`) → expect `Python 3.12.x`; then `pip install pandas numpy matplotlib jupyter streamlit`.
4. **File** → **New File**, name it `hello.py`, type `print("Hello, world!")`, save, click the ▷ Run button top-right.
   ✔ **Hello, world! appears in the terminal panel**

If something goes wrong — the usual suspects

Colab: "not available for your organisation"
Your university Google account blocks it. Use a personal Gmail.

Colab: "Cannot connect to runtime" / spinning forever
Runtime → Disconnect and delete runtime, run again. Turn off ad blockers and VPN.

GitHub verification email never arrives
Check spam; wait five minutes; add a second email under Settings → Emails.

Stuck? Here is the order of operations

1. Paste the exact error message plus one line on what you were doing into your AI assistant from step 5, and follow its suggestion. Nine times out of ten that is the end of it. Still stuck after two honest tries? Search the exact error text on the web.
2. Then email limor.gultchin@gmail.com with the step you were on, a screenshot of the error, what the assistant suggested, and Mac or Windows — **before Day 1** so there is time to fix it.

Getting stuck and then unstuck is not a detour from learning to code — it *is* learning to code.

Done? Tick every box before Day 1

- ☐ Hello, world! ran in Colab; notebook saved as hello.ipynb
- ☐ GitHub account + two-factor + hello-workshop repository
- ☐ 00_setup_check.ipynb ends with "You're ready ✔"
- ☐ Pre-reading done and survey submitted
- ☐ Three package versions printed, no red text
- ☐ hello.ipynb saved from Colab into hello-workshop
- ☐ AI assistant account works and explained the NameError
- ☐ Laptop charged, charger packed, both logins known